

Task 1 — System Analysis Task

Library Management System - COMPLETED

Entity 1: Member

Property Name	Data Type	Source Tag
MemberId	int	System Generated
Name	string	User Input
Email	string	User Input
PhoneNumber	string	User Input
RegistrationDate	DateTime	System Generated
IsActive	bool	Default Value

Entity 2: Book

Property Name	Data Type	Source Tag
BookId	int	System Generated
Title	string	User Input
Author	string	User Input
Genre	string	From List
TotalCopies	int	User Input
AvailableCopies	int	Calculated

Entity 3: Loan

Property Name	Data Type	Source Tag
LoanId	int	System Generated
MemberId	int	From List
BookId	int	From List
BorrowDate	DateTime	System Generated
DueDate	DateTime	Calculated
ReturnDate	DateTime	User Input
IsReturned	bool	Calculated

Entity 4: Reservation

Property Name	Data Type	Source Tag
ReservationId	int	System Generated

MemberId	int	From List
BookId	int	From List
ReservationDate	DateTime	System Generated
QueuePosition	int	Calculated
NotificationSent	bool	Default Value

Entity 5: Fine

Property Name	Data Type	Source Tag
Fineld	int	System Generated
LoanId	int	From List
DaysOverdue	int	Calculated
FineAmount	decimal	Calculated
IsPaid	bool	Default Value
PaymentDate	DateTime	User Input

Entity Relationships:

Member → **Loan**: One-to-Many. A member can have multiple loans over time.

Book → **Loan**: One-to-Many. A book can be loaned multiple times.

Loan → **Fine**: One-to-One. A loan generates at most one fine record.

Member → **Reservation**: One-to-Many. A member can reserve multiple books.

Book → **Reservation**: One-to-Many. Multiple members can reserve the same book (forming a queue).